



TASK 05-50-03-200-804-A

EFFECTIVITY: ALL

1. Off-Temperature Envelope Flight - Inspection/Check

A. General

- (1) This task contains the necessary actions for aircraft operated outside the flight envelope temperature given in the AFM.
- (2) This section gives the necessary actions after a flight envelope temperature exceedance is reported by crew, or detected by flight data analysis.
- (3) The temperature flight envelope is presented in the AFM - Section 2 - Limitations, and is reproduced in the Temperature Allowance Chart [Figure 619](#), together with the thresholds and exceedance zones.
- (4) If the temperature flight envelope given in [Figure 619](#) is exceeded during flight, do the steps at the Job Setup [SUBTASK 160-002-A](#).
- (5) If the temperature flight envelope given in [Figure 619](#) is exceeded for more than 10 minutes in the same flight, contact RTS.
- (6) The operator can do the off-temperature envelope flight analysis, but if you choose to send the data for EMBRAER analysis, it is recommended that you send the QAR file.
- (7) If the landing was below -45° C the AMM TASK 05-50-03-200-801-A/600 is applicable.
- (8) This maintenance technician is necessary to do this task:
 - 1 - Electrical/Avionics Systems

B. References

REFERENCE	DESIGNATION
AMM TASK 05-50-03-200-801-A/600	Hard Landing And Off Temperature Envelope Landing - Inspection/Check
AMM TASK 20-00-00-910-801-A/200	Aircraft Maintenance Safety Procedures - Standard Procedures
AMM TASK 32-49-11-210-801-A/600	Brake Assembly Wear-Pin (Fast check - Wheel installed on aircraft) - General Visual Inspection

C. Job Set-Up

SUBTASK 160-002-A

- (1) If the temperature flight envelope is exceeded in a cold condition areas as per [Figure 619](#) during the flight or when flight colder than -40°C under (10000 ft), do the steps that follow:

NOTE: The operator can do the analysis of the flight operation data, but if you choose to send the data for an EMBRAER analysis, it is recommended that you send the QAR file.
- (2) Determine the coldest Total Air Temperature (TAT) experienced during the cold condition exceedance in [Figure 619](#).



(a) Total Air Temperature (TAT) can be obtained through QAR file (total AirTemperature-1), or;

(b) Do the Total Air Temperature (TAT) calculation, as follow:

- $TAT = SAT [1 + (0.2 M^2)]$, where;
NOTE: The temperature must be converted to K (Kelvin) for calculation, and then changed to °C (Celsius) for verification.
- $TAT = \text{Total Air Temperature [K]}$
- $SAT = \text{Static Air temperature [K]}$
- $M = \text{Mach Number}$

(3) If the calculated engine, skin and brake temperature is below the thresholds, there is no further action and Embraer do not need to be contacted.

D. Skin temperature check/inspection

SUBTASK 160-004-A

- (1) Do the skin temperature calculation [SUBTASK 160-002-A](#).
- (2) Is the calculated skin temperature -54 °C or colder?
 - (a) YES - Contact the RTS and supply QAR data for analysis and recommendations.
 - (b) NO - No more actions for skin are necessary.

E. Engine temperature check/inspection

SUBTASK 160-005-A

- (1) Do the engine temperature calculation [SUBTASK 160-002-A](#).
- (2) Was the total air temperature colder than -54°C?
 - (a) YES - Do the engine inspection (Contact RTS)
 - (b) NO – No more actions are necessary

F. Brake fluid leakage check/inspection

SUBTASK 200-007-A

- (1) Do the skin temperature calculation [SUBTASK 160-002-A](#).
- (2) Was the total air temperature colder than -40°C?

NOTE: For aircraft with extended operational envelope the threshold is -45°C

- (a) YES - Do a GVI in the brakes ([Figure 620](#)).

NOTE: Do GVI with wheels installed on aircraft.

If any brake leakage, do the (AMM TASK 32-49-11-210-801-A/600).



(b) NO – No more actions are necessary

G. Job Close-Up

SUBTASK 000-002-A

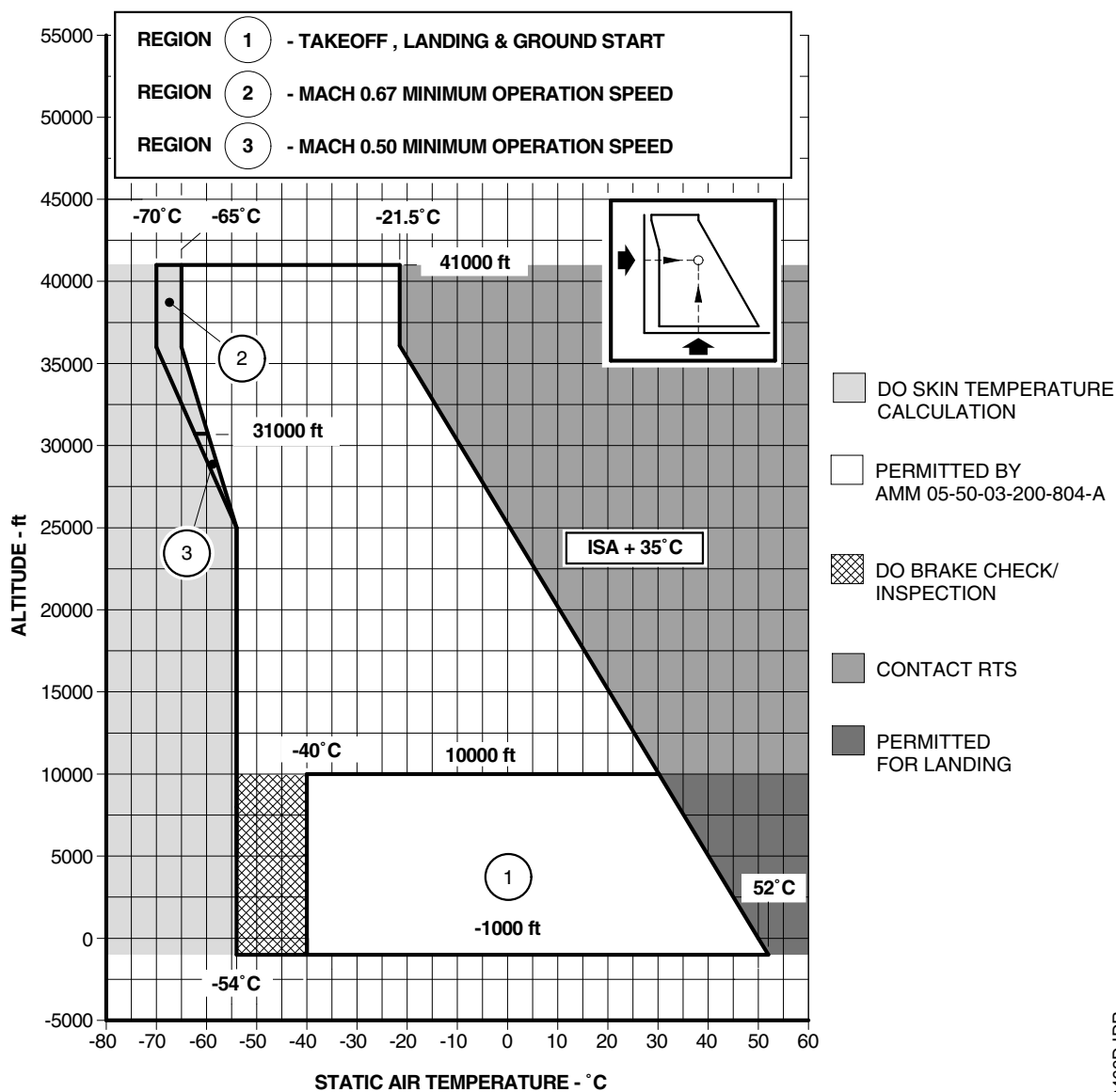
(1) Put the aircraft back to its initial condition (AMM TASK 20-00-00-910-801-A/200).



EFFECTIVITY: ALL

Temperature Allowance Chart

Figure 619



1 FOR AIRCRAFT WITH EXTENDED OPERATIONAL ENVELOPE THE THRESHOLD IS -45°C

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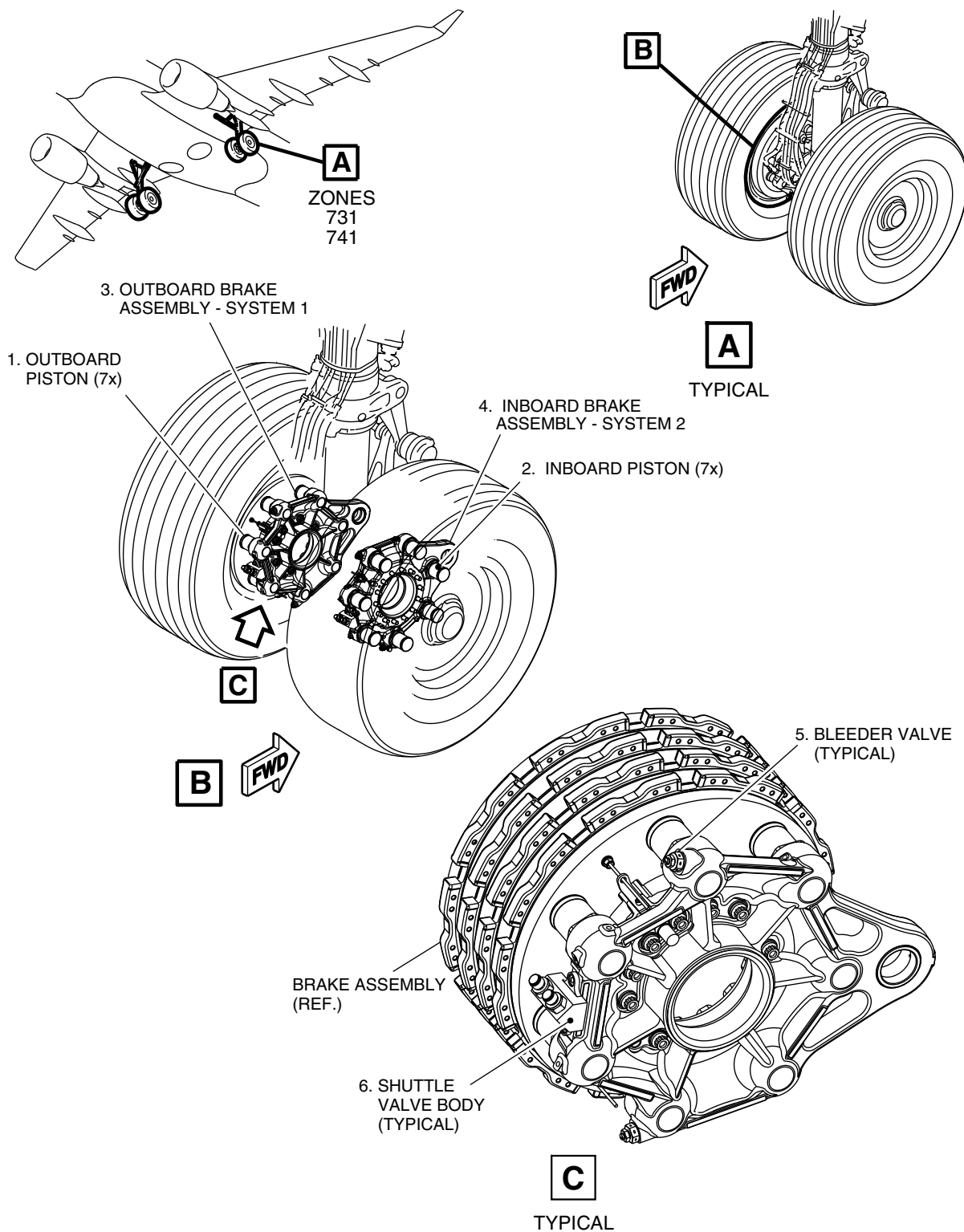
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EFFECTIVITY: ALL

Break fluid leakage check/inspection

Figure 620



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